

Jesse Baugh

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EDUCATION

Bellevue College - *Bachelor of Science, Computer Science*

Expected Graduation: Spring 2028

RELEVANT COURSEWORK

- Fundamentals of Computer Science I-III, Data Structures, Database Systems, Calculus I-III, Linear Algebra.

TECHNICAL SKILLS

- Languages:* Java, JavaScript, TypeScript, Python, C++, SQL, HTML
- Tools & Frameworks:* Git, Node.js, Express, Docker, JUNIT
- Concepts:* API Design, Data Pipelines, DAG-based Workflow Execution, OCR Pipelines
- Human Languages:* English & Spanish

PROFESSIONAL EXPERIENCE

CodeDay Labs - Open-Source Software Engineering Intern

June – August 2026

Maxun | TypeScript, Node.js, Playwright, Tesseract.js, PaddleOCR

- Expanded Maxun's OCR capabilities by integrating Tesseract.js and PaddleOCR for JPG/PNG ingestion, improving non-text document extraction coverage and enabling structured Markdown/HTML output for downstream tools.
- Identified reliability issues in the existing extraction pipeline and engineered a Prompt-Wrapping Orchestrator that enforced strict JSON boundaries, reducing malformed outputs by ~30% and enabling deterministic extraction across PDF, DOCX, XLSX, and CSV files for both cloud and local LLMs (e.g., ollama:llama3.2).

Full-Service Restaurants, Multiple Locations - Restaurant Supervisor/Lead Bartender

2018 – 2025

- Configured and updated TOAST POS system databases to reflect seasonal menu transitions, ensuring accurate pricing, inventory tracking, and seamless order routing.
- Led shift-level operations across several high-volume restaurants, coordinating workflows between front-of-house and kitchen teams to maintain smooth, efficient service.

PROJECTS

Vector Workflow Engine | *TypeScript, Node.js, Express, OpenAI API*

A backend developer tool designed to parse and execute LLM-generated plans using dependency-aware actions.

- Built a dependency-aware execution engine leveraging Directed Acyclic Graphs (DAGs), cycle detection, and topological sorting to compute safe parallel and sequential execution layers.
- Engineered a fault-tolerant validation subsystem featuring an LLM-driven autofix mechanism to automatically repair malformed JSON steps, missing fields, and mid-plan crashes.
- Developed an optimization pipeline that improves computational efficiency by removing redundant steps, deduplicating file writes, and deterministically grouping independent tasks for concurrent execution.
- Implemented a robust tool registry and execution pipeline supporting scalable file operations, memory storage, skip/retry logic, and structured telemetry.

Seattle Care-Access | *Python, Pandas*

A platform designed to streamline access to healthcare resources by connecting users with local care providers and support services in Seattle, WA.

- Implementing core features such as user authentication, data intake workflows, and provider matching logic.
- Collaborating with faculty mentors to refine system architecture and ensure scalability and accessibility.
- Built reproducible data pipelines using Python and Pandas; version-controlled with Git/GitHub.